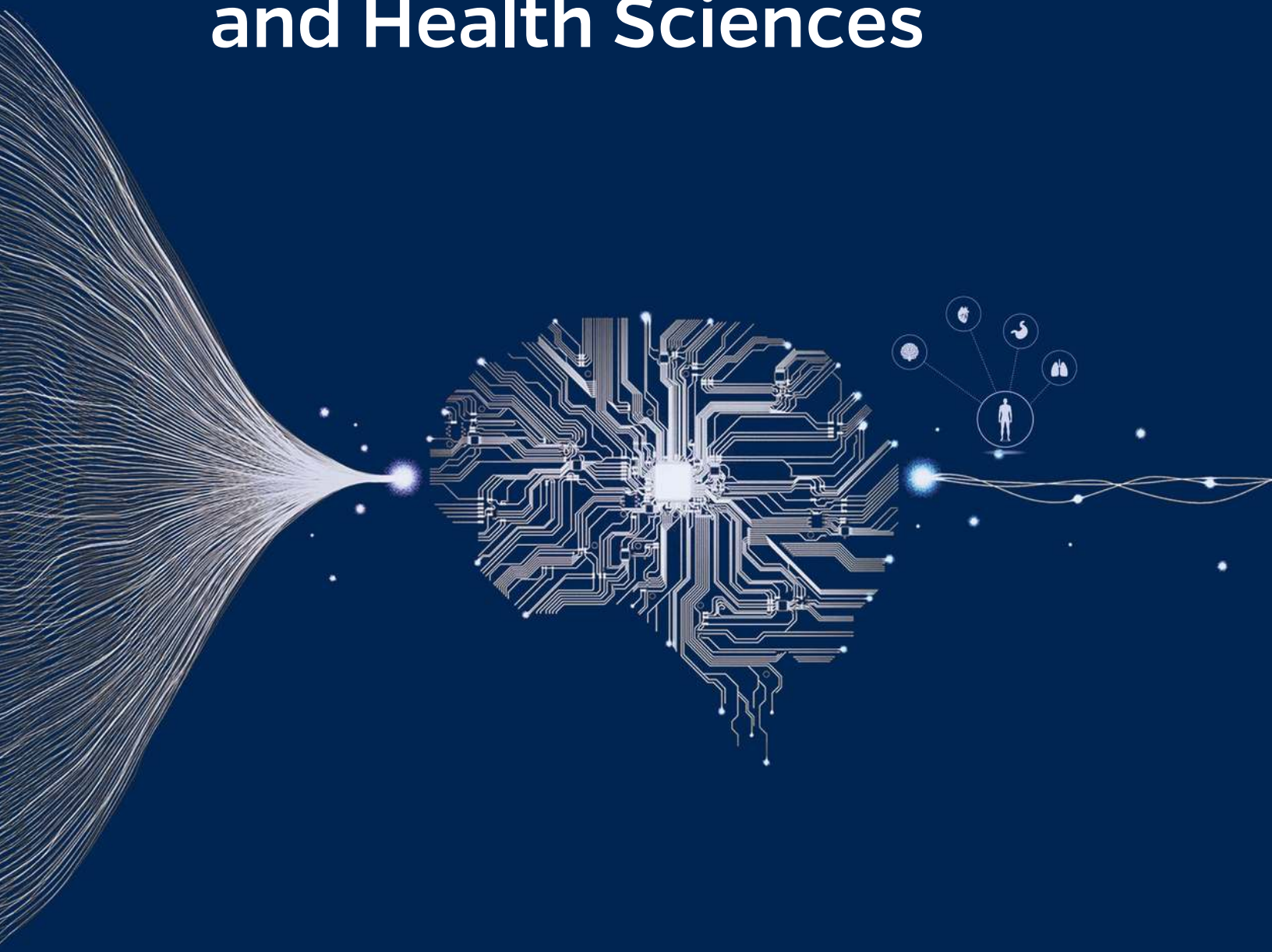
M.S. Program

AIBHS | Artificial Intelligence in Biomedical and Health Sciences

AIin Biomedical and Health Sciences



UF | College of Medicine UNIVERSITY of FLORIDA

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AIBHS | Artificial Intelligence in Biomedical and Health Sciences

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AI at UF

AIBHS IS A TRANSDISCIPLINARY PROGRAM DESIGNED TO USE TRUSTWORTHY AI TO AUGMENT HUMAN-CENTERED HEALTHCARE AND TRANSFORM CRITICAL AND ACUTE CARE MEDICINE.

UF has developed new AI degrees accompanying over 230 AI courses.

In 2020, NVIDIA and Chris Malachowsky, founder of NVIDIA and alumnus of UF, gifted UF $50 million and HiPerGator: one of the fastest supercomputers in academia. Prompted by this generous donation, UF launched its artificial intelligence (AI) Initiative to make the university a national leader in AI research, teaching, and economic development. Since this launch, UF has developed new AI degrees accompanying over 230 AI courses. **The M.S. in Artificial Intelligence in Biomedical and Health Sciences (AIBHS) is part of this initiative.**

Housed in the college of medicine's Intelligent Clinical Care Center (IC3), AIBHS is a transdisciplinary program designed to use trustworthy AI to augment human-centered healthcare and transform critical and acute care medicine. Through the partnership between the UF colleges of medicine and engineering, the IC3 is the first transdisciplinary program in the nation to study AI in critical and acute illnesses, bringing together medicine, engineering, education, IFAS, pharmacy, and law.

Aligning with the goals of the IC3, AIBHS is a collaborative enterprise uniquely immersing students in the various facets of biomedical and clinical AI, providing a cohesive understanding to accelerate biomedical advancements and improve patient care.

About the Program

Learn to utilize AI in real-world biomedical & clinical settings

AIBHS, conveniently co-located in Jacksonville and Gainesville, is a transdisciplinary program designed to teach students to use trustworthy AI to augment human-centered healthcare and transform critical and acute care medicine. AIBHS emphasizes hands-on learning and first-hand experience developing and implementing artificial intelligence technologies to solve real-world biomedical challenges and revolutionize patient care.

AIBHS Students can look forward to:

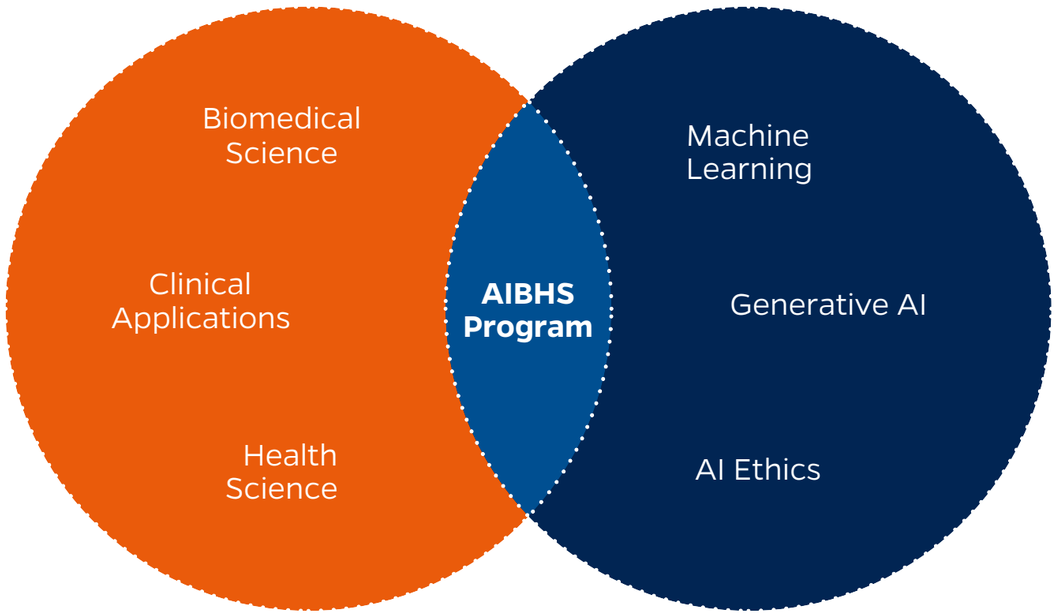
Hybrid Learning
(Online & Face-to-Face)

Networking & Collaboration
(Jacksonville and Gainesville)

Dual Mentorship in AI & Biomedical or Clinical Science

Transdisciplinary Training

Combining
AI & Biomedical Sciences



Traditional disciplines often lack a nuanced focus on biomedical and health-focused AI, while data science generally misses an emphasis on applications for translational biomedical domains and clinical settings.

AIBHS offers community-embedded experiential training, where biomedical and clinical expertise is matched with AI technical, ethical and legal expertise. AIBHS is a holistic program, enabling students to practice and apply skills directly into translational biomedical research and clinical settings.

Learn how to accelerate healthcare advancements and improve patient care with an M.S. degree in Artificial Intelligence in Biomedical and Health Sciences.

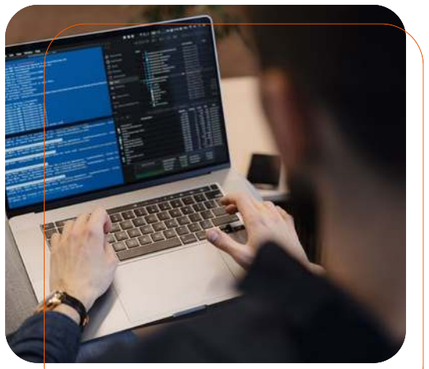
Curriculum

30 credit coursework curriculum equipping students with the knowledge and skills to contribute impactfully to the emerging intersection between AI, biomedical research, and clinical care.

All types of students welcome:

Working professionals, traditional graduate students, those pursuing other professional or graduate degrees, etc. are encouraged to apply

FULL-TIME STUDENTS CAN GRADUATE IN 1 & 1/2 YEARS



Recommended Background

Foundational biological or medical knowledge
(i.e. biology degree or clinical practice experience)
and/or strong computational background
(i.e. computer science, statistics, or engineering)
Python or R familiarity recommended (optional)

Core courses:

Fundamentals of AI in Medicine I
Fundamentals of AI in Medicine II
Biostatistics for AI
AI Ethics and Alignment in Health

Experiential Learning courses:

AI Design Studio in Health I
AI Design Studio in Health II
Clinical AI Design Studio I
Clinical AI Design Studio II

Experiential Electives:

AI in Medical Image Analysis
Applied Generative AI in Medicine
AI for Clinical Decision Support
AI-Powered Drug Discovery
Supervised Research

Design Studios:

Via AI and Clinical Design Studios, students will learn from AI technical experts and biomedical and clinical researchers.

Job Outlook:
AI & Biomedical Science

$95,070

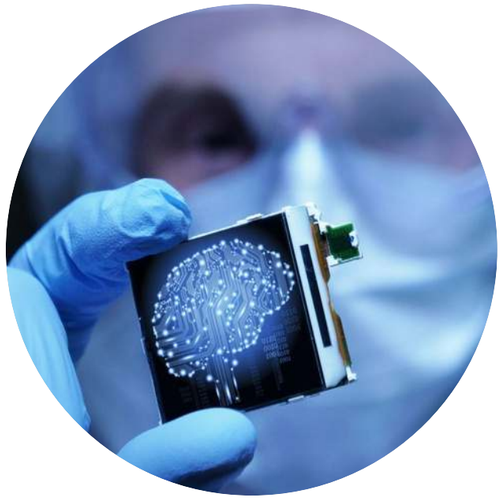
Median salary in Florida

7-35%

Faster than average projected national job growth

$99,213

Median national salary



AIBHS graduates are qualified for a myriad of careers:

- Data Scientist
- Medical & Health Services Manager
- Database Architect
- Technical Writer
- Computer & information Research Scientist
- Health Information Technologist

Dual Mentorship for Clinical & AI Expertise

AIBHS's transdisciplinary curriculum integrates machine learning, generative AI, and AI ethics with biomedical research, medical practice, and clinical workflows.

Dual Mentorship

Students have two mentors: a technical AI expert and a clinical expert

Mentors will train students to apply technical knowledge to effect concrete, real world biomedical and clinical improvements.

AI and Clinical Design Studios to Enhance Learning

AIBHS's transdisciplinary curriculum integrates machine learning, generative AI, and AI ethics with biomedical research, medical practice, and clinical workflows.

Signature Design Studios

AI Design Studios:

Domain experts guide students through the design and application of AI systems dedicated to resolving real world medical problems fairly

Clinical Design Studios:

Students rotate with esteemed clinical or basic faculty, acquiring direct experience in clinical settings or biomedical research labs

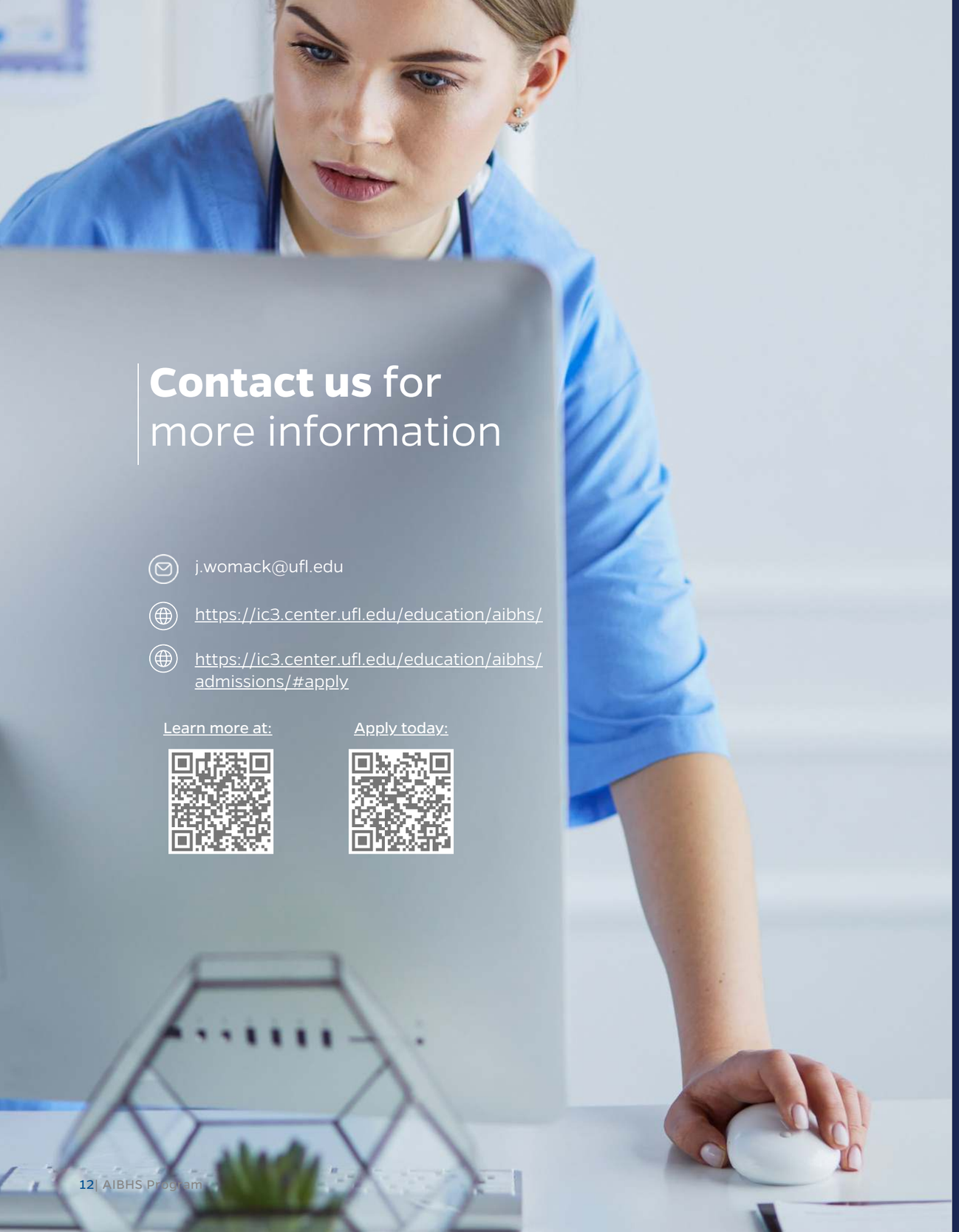
AIBHS

Meet our **Program Director**

Azra Bihorac | MD, MS, FCCM, FASN

Program Director

Senior Associate Dean for Research at the University of Florida College of Medicine; R. Glenn Davis Professor of Medicine, Surgery and Anesthesiology; Director, Intelligent Clinical Care Center; Director, Precision and Intelligent Systems in Medicine Partnership



Contact us for more information

j.womack@ufl.edu

https://ic3.center.ufl.edu/education/aibhs/

https://ic3.center.ufl.edu/education/aibhs/admissions/#apply

Learn more at:

Apply today:

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Learn to utilize AI in real-world biomedical & clinical settings

j.womack@ufl.edu

1889 Museum Road
Gainesville, FL 32611

Learn more at:

Apply today:

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